

Nicholas Occhipinti

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Charlotte, NC 📍

Experienced professional delivering insights through Data Science, Geospatial Technologies and Solution Design to optimize resource allocation, inform strategic decision-making, and drive innovative solutions.

EXPERIENCE

Urban Institute at UNC Charlotte, Charlotte, NC— *Data Scientist*

JUNE 2021 - PRESENT

- Work on the Charlotte Regional Data Trust where I securely integrate individual-level administrative data across agencies in the Charlotte–Mecklenburg region. By linking records through a robust, privacy-preserving system, the Data Trust is able to repurpose administrative data for research, evaluation, and planning.
- Co-authored the inaugural [State of Our Data](#) report which highlights different statistics and visualizations about people in the Charlotte–Mecklenburg region that are in the data we maintain. Given that much of the data reflects marginalized or underserved communities, the report aims to equip changemakers with the tools, systems, and insights needed to address regional social, economic, and environmental challenges.
- Developed an entity resolution process that links individuals across datasets using names and birthdates, enhanced with demographic information. Created with a synthetic dataset containing fake names, birthdates, addresses, gender, and race (assigned via Bayesian Improved Surname Geocoding). Variations and noise (e.g., name misspellings, twins) are introduced to simulate real-world inconsistencies. The matching process involves five key steps: preprocessing (standardizing formats, removing noise, and applying Soundex phonetic encoding), indexing (narrowing candidates using birthdate and Soundex name codes), comparing candidate pairs (using string similarity and field weights), classifying matches with a Random Forest model, and deduplicating matches and assigning anonymized IDs.
- Created an Airflow orchestration pipeline that processes data deposits from over 12 partners, handling various formats and cleanliness levels by unzipping files, converting them to CSV, designing database schemas and loading the data into a database. It includes automated data cleaning based on field and deposit types, record comparison for inserts or updates, and comprehensive logging for error tracking. The process will also generate stats for every field which feeds into visualization database tables used in AWS Quicksight dashboards.
- Built a web mapping application in Angular that leverages the ArcGIS JavaScript API and hosted feature layers from ArcGIS Online to visualize regional trends. The tool presents indicators across demographic, economic, and educational categories at the Census Tract level for a 14-county region supported by UNC Charlotte's Urban Institute. Designed for use by community members, researchers, and decision-makers, it enables users to explore how areas are changing over time. Built-in analysis tools include correlation analysis between indicators and Moran's I to identify spatial clustering patterns.

UNC Charlotte, Charlotte, NC— *GIS Developer*

NOVEMBER 2017 - JUNE 2022

- Provided GIS support for a variety of departments across campus such as Facilities Management, Police and Parking Services.

- Maintained a Campus web map built in Angular that integrated with different API's to show live tracking of buses, bike share dock availability and available parking deck space counts.
- Administered the Campus ArcGIS Server and Portal applications that are used to distribute web maps to view and edit data in the field.
- Develop Tableau visualizations and dashboards and also provided training for other employees on how to use Tableau to visualize their data.

New York City Department of Education, Brooklyn, NY— *GIS Developer*

APRIL 2017 - AUGUST 2017

- Created and maintained spatial datasets for the New York City school system, consisting of over 1,800 schools and 1.1 million students.
- Maintained an ASP.NET MVC mapping application and related ArcGIS map services and automated GIS tasks using Python.

Mott MacDonald, Iselin, NJ— *GIS Specialist*

JUNE 2007 - MARCH 2017

- Developed Python scripts for a variety of uses such as FTP image transfer, spatial analysis, creating map books and developing geoprocessing services.
- Developed an ASP.NET MVC mapping application to view and edit data, that played a pivotal role in acquiring new clients.
- Architected a document management system that linked documents to spatial assets in a web-based GIS application.
- Administered a production GIS Server environment to host data for approximately 25 clients.

EDUCATION

University of North Carolina at Charlotte, Charlotte, NC

Master's Degree - Data Science and Business Analytics

September 2020 - December 2021 - 4.0 GPA

University of North Carolina at Charlotte, Charlotte, NC

Graduate Certificate - Data Science and Business Analytics

January 2019 - December 2019 - 4.0 GPA

Oregon State University, Corvallis, OR

Graduate Certificate - Geographic Information Science

January 2012 - May 2014 - 3.93 GPA

Kean University, Union, NJ

Bachelor of Science Degree - Computer Science

September 2001 - May 2004